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CS4055: Networks and communications
W10

UTEC
UNIVERSIDAD DE INGENIERÍA Y TECNOLOGÍA

LAB3 TRANSPORT LAYER

TCP

The present evaluation will be received **only if the checkpoints of all the questions in the LABORATORY SESSION have been completed** with the course teacher.

For the checkpoints presentation, **have a doc or text file** where you present the answer to the questions (**including screenshots if necessary**). **Do not forget to save your tracefile.**

I. The submission must consist of:
   A. **Introduction** (include introductory paragraph):
   B. **Theoretical Framework** (definitions with references).
   C. **State of the Art** (research papers, articles, theses, pages with references).
   D. **Development of the experience** (including discussion and correct figure formatting)
   E. **Conclusions and recommendations** (at least one for each experience carried out).
   F. **References** (IEEE format)

II. The submission must be presented in **LaTeX** or Docs format (**pdf file**).

III. **Upload only the PDF of the report to Canvas**, use a link to the repository, include step-by-step photos of each experience and answer checkpoints where relevant, plus codes if they have been developed.

IV. **Only one member of the group will upload the report**.

V. **Do not forget to list the names of all members**.

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LAB3 TCP P2

5. Python TCP Server-client implementation.

5.1. Implementation Task

Develop a basic TCP client-server application in Python. Your code must successfully establish a connection, triggering the underlying TCP three-way handshake.

Recommended Resources:

- https://rico-schmidt.name/pymotw-3/socket/tcp.html
- https://realpython.com/python-sockets/ *(Note: Ensure you are using Python 3, as Python 2 is deprecated).*

Requirements:

- **Port Configuration:** Configure your server to listen on a custom port number using the last 5 digits of your Student ID or DNI. *(Note: The maximum valid port number is 65535. If your 5 digits form a larger number, use only the last 4 digits).*
- **Packet Capture:** Use Wireshark to capture the network traffic generated between your client and server during execution.

5.2. Analysis Questions

Answer the following questions based on your implementation:

1. Which network interface did you select in Wireshark to capture the traffic generated by your Python code? (Explain why).
2. How did you configure the server to use your Student ID/DNI as the listening port number?
3. Which specific Python socket functions triggered the OS to establish the TCP three-way handshake between the client and the server?
4. What steps did you follow to run both the server and client programs simultaneously on your system?

CHECKPOINT 4

6. Sending a File over TCP

6.1. Implementation Task

Building on the previous laboratory session, extend your program into a file transfer application capable of sending a .txt file from the client to the server.

Requirements:

- **Target File:** Use the provided text file for this exercise: **RFC_TCP.txt**.
- **Packet Capture:** Capture the complete data transmission process using **Wireshark**.

6.2. Deliverables & TCP Traffic Analysis

Code: Present the resulting Python source code for both the client and server implementations.

Demonstration: Show the live Wireshark trace of the file transfer to your instructor.

Questions:

1. How many TCP segments were generated in total during the transmission of the .txt file?
2. Inspect the initial three-way handshake (SYN packets). What Maximum Segment Size (MSS) was negotiated? How does this value mathematically relate to your interface's MTU (Maximum Transmission Unit)?
3. What patterns can you identify in the TCP Sequence Numbers and Acknowledgment Numbers as the file data is being transferred?
4. Were all packets delivered successfully, or did you observe any TCP error recovery mechanisms (e.g., TCP Retransmission, Fast Retransmission, Dup ACKs) in Wireshark?

6.3. Scenarios: MTU, ICMP

Temporarily lower the MTU of your active network interface (e.g., to 1000 bytes) using your operating system settings or command line, and repeat the TCP file transfer.

Questions:

5. How did the negotiated MSS change in the new TCP handshake?
6. How did this MTU reduction affect the total number of captured segments?
7. *(Optional)* If you force the Don't Fragment (DF) bit, do you capture any **ICMP "Fragmentation Needed"** or "Destination Unreachable" packets? Explain why they appear.

CHECKPOINT 5